

Unified Multi-Environment Cyber Threat Detection & Response Platform

Real-Time SIEM • MITRE ATT&CK • ML Anomaly Engine • Google Gemini 2.0 Flash • SHA-256 Audit Ledger

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Objective

To develop an autonomous multi-environment Security Operations Center (SOC) platform that unifies cross-cloud telemetry, maps threats to MITRE ATT&CK, enforces cryptographic SHA-256 audit integrity, and provides real-time AI analyst guidance using **Google Gemini 2.0 Flash**.

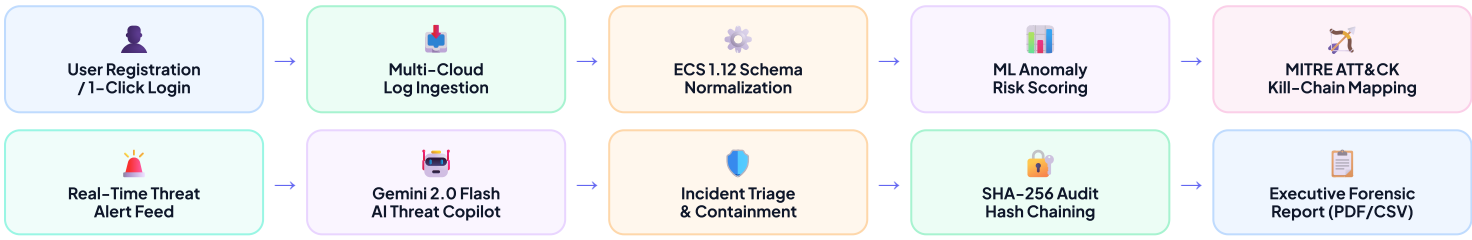


About Project

A production-grade cloud-native SOC platform built with React 18, FastAPI, and WebSocket streams. It normalizes distributed logs under Elastic ECS 1.12, computes composite risk scores (0-100), offers 1-click registered user access, and provides conversational threat triage via Google Gemini AI.



Project Workflow

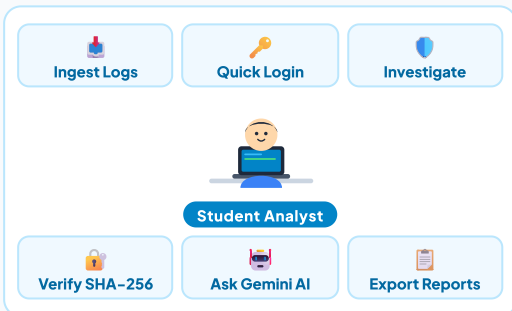


Key Features

- ✓ Multi-Cloud Telemetry (AWS, Azure, Linux)
- ✓ ECS 1.12 Auto-Normalization & Parsing
- ✓ ML Anomaly Scoring Engine (0-100 scale)
- ✓ MITRE ATT&CK v14 Kill-Chain Alignment
- ✓ Gemini 2.0 Flash Multimodal AI Threat Analyst
- ✓ Conversational SOC Threat Q&A & Attribution
- ✓ Human-in-the-Loop Analyst Decision Workflows
- ✓ SHA-256 Cryptographic Tamper-Proof Audit
- ✓ 1-Click Registered User Login Recognition
- ✓ Executive Forensic PDF & CSV Audit Exports



Student View & Output Flow



"As a student security operator, working with raw cloud logs used to feel overwhelming. This platform gave me an intuitive experience: I register my account, pick my profile from the login screen, upload messy multi-cloud logs, and immediately see clear threat classifications instead of endless text. Whenever a critical alert triggers, I simply ask the integrated Gemini 2.0 Flash AI to explain the attacker's motive and give me the exact containment commands. The SHA-256 audit chain guarantees my forensic evidence is mathematically unalterable."

OUTPUT VIEW FLOW (DELIVERABLES)



Tools Used

- VS Code
- Git & GitHub
- Vite CLI
- Postman / Swagger
- Vercel Cloud
- Docker



Technologies Used

- FRONTEND:
 - React 18
 - TypeScript 5
 - Tailwind CSS
 - WebSocket (RT)
- BACKEND:
 - Python FastAPI
 - SQLite / DB
 - Redis Queue
 - JWT Token Auth
- AI & THREAT INTELLIGENCE:
 - Google Gemini 2.0 Flash
 - MITRE ATT&CK v14
 - AbuseIPDB API
 - SHA-256 Crypto

2026 CASE STUDY Operation Cyber Guardian: Critical Multi-Cloud Infrastructure Infiltration (2026)

Gemini 2.0 Flash AI Advantage

Parameter	✗ 2026 Real Breach (Legacy Disconnected SOC)	✓ Present Unified Platform (With Gemini 2.0 AI)
Detection Speed (MTTD)	Undetected for >180 days across hybrid cloud & Linux servers.	Automated detection in <5 minutes via ECS 1.12 normalizer.
Attack Correlation	Disconnected tools failed to trace lateral movement across hypervisors.	Auto-maps MITRE T1078 (Credentials) & T1021 (Lateral Movement).
Forensic Audit Trail	Attackers deleted Linux syslog files, destroying evidence trails.	SHA-256 cryptographic chain immediately flags log tampering.
Analyst Response (MTTR)	Analysts overwhelmed by 100k+ alerts; manual triage took months.	Gemini 2.0 Flash provides mitigation script; resolved in 4 hours.



GitHub Repository

github.com/Pravallika2025/unified-threat-platform



Live Cloud Deployment

unified-threat-platform.vercel.app